

Course Code	21DCS201P	Course Name	DESIGN THINKING AND METHODOLOGY		Course Category	S	ENGINEERING SCIENCES												L	T	P	C
																			1	2	0	3
Pre-requisite Courses		Nil	Co-requisite Courses	Nil	Progressive Courses		Nil															
Course Offering Department		SRM Innovation and Design Centre		Data Book / Codes / Standards	Nil																	
Course Learning Rationale (CLR):		The purpose of learning this course is to:			Program Outcomes (PO)										Program Specific Outcomes							
CLR-1 :	designed to explores mindset, skill set and toolset associated with design			1	2	3	4	5	6	7	8	9	10	11	12	PSO-1 PSO-2 PSO-3						
CLR-2 :	designed to work with guided applications to framing and solving problems from the perspectives of both business and engineering writing			Engineering Knowledge	Problem Analysis	Design/development of solutions	Conduct investigations of complex problems	Modern Tool Usage	The engineer and society	Environment & Sustainability	Ethics	Individual & Team Work	Communication	Project Mgt. & Finance	Life Long Learning							
CLR-3 :	exposing students diverging to generate solutions and converging to select among them			-	-	2	-	-	-	-	-	-	-	-	-							
CLR-4 :	design methods to create concept generation methods, concept selection methods, imagining alternative futures			-	-	-	3	2	-	-	-	-	-	-	-							
Course Outcomes (CO):		At the end of this course, learners will be able to:																				
CO-1:	learn and understand technology design concepts			-	-	2	-	-	-	-	-	-	-	-	-	-						
CO-2:	learning mindset, skillset and toolset associated with design			-	-	-	3	-	-	-	-	-	-	-	-	-						
CO-3:	identify the best solutions and converging to select among them			-	2	-	-	-	-	-	-	-	-	-	-	-						
CO-4:	understand concept generation methods, concept selection methods, imagining alternative futures			-	-	3	2	-	-	-	-	-	-	-	-	-						
Unit-1 - Introduction to Design Methodology																	9 Hour					
Design Frameworks - Engineering Design Problem Solving - Developing Design Solutions - Making Design Solutions - Evaluating Design Solutions - Project Introduction and Team formation -Stakeholder Map - Brainstorming																						
Unit-2 - Prototyping																	9 Hour					
Prototyping Planning - Concept Refinement and Storyboard - Envisioning Future - Conceptual design - Creative Matrix, Morphological Synthesis, Concept Poster - Basic Presentation - Project Discussion with Teaching Team – Usability Test Demo																						
Unit-3 - User Assessment																	9 Hour					
Usability Test - Understanding Users - Learning about Customer - Clustering & Abstract Laddering - User Testing - Project Discussion with Teaching Team																						
Unit-4 - Value Proposition Design																	9 Hour					
Value Proposition Design and Mapping, - Prototyping & Competitor Study - Competitors / Complementor's Map - Design Methodologies - Capital Budgeting: Risk Analysis with Scenarios - Project Discussion with Teaching Team - Project Mini-showcase - Project Discussion with Teaching Team																						
Unit-5 - Business Model																	9 Hour					
Business Model Canvas - Business pitch - Pitching strategies - IP and Partnerships - Forecasting Financial Statements - Project Discussion with Teaching Team - Final Project Presentation - Design Showcase																						
Learning Resources	1. Dr. R. Thomas Wright, Dr. Greg J. Strimel, and Dr. Michael E. Grubbs Foundations of Engineering & Technology, 7th Edition																2. Ikhlagsidhu , Innovation Engineering; a practical guide to creating anything new					